

Sameer Khan

SENIOR ML ENGINEER · DOCUMENT AI & MULTIMODAL · LLM / RAG · MLOPS ON AWS

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DOMAIN EXPERTISE

- Mortgage & Loan File Documents
- Retail Banking — Arabic / English
- Identity Verification & KYC

TECHNICAL SKILLS

ML / Deep Learning
PyTorch · PyTorch Lightning · TensorFlow / Keras · HuggingFace Transformers · CNN · ResNet50 · Swin / ViT · Donut · Multimodal Fusion · BiLSTM · NER · LoRA / QLoRA / PEFT · SFT · Instruction Tuning · RLHF-DPO · fp16 / bf16

Distributed Training
Horovod · DDP · TorchDistributor · NCCL · Multi-GPU · Spark / PySpark · Databricks

Document AI / CV
Document Classification · Information Extraction · OCR & OCR-Free · Deduplication · Signature Verification · Object Detection · Tesseract · AWS Textract · TF Object Detection · OpenCV

LLM / RAG
RAG · FAISS · Dense Retrieval · Okapi BM25 · Hybrid Retrieval · RRF · Cross-Encoder Reranking · JSON Schema Outputs · Selective Prediction · Confidence Calibration · Llama 3 · Claude API

Serving / MLOps
AWS SageMaker · Real-Time Endpoints · Autoscaling · Model Registry · TF Serving · ONNX · Jenkins CI/CD · Docker · MLflow · CloudWatch

Production Practices
Distributed Training · Mixed Precision · Model Export & Optimisation · Selective Prediction · Threshold Calibration · Incident RCA · Release Governance · Change Control · Cross-Account Promotion · Rollback

Languages / Tools
Python · SQL · Git · Linux / bash

EDUCATION

B.Tech, Computer Science Engineering
TKR Engineering College (JNTU) Hyderabad · 2016

Languages
English · Hindi

PROFILE

Senior Machine Learning Engineer with **9 years in Document AI** — computer vision, NLP and multimodal deep learning applied to one problem across three roles, turning scanned documents into structured, reliable data. Currently own ICE's production mortgage page classifier: **643 classes at ~1.7M requests/day** on SageMaker with **99.99% success**, and I proposed and built its **OCR-free transformer successor**. Also built near-duplicate detection across a 7.1M-page corpus, and a bank's document-digitisation platform as its first ML engineer. Hands-on across the full lifecycle — architecture, distributed training, the serving path, and production incident response. Targeting senior individual-contributor roles with architecture ownership.

EXPERIENCE

- ICE Data Services** — Hyderabad **Nov 2023 – Present**
Senior ML Engineer
- Own ICE's production mortgage document classifier end to end. **643 classes**, trained on **2.8M pages** from 49 encrypted lender sources. It serves **1.7M requests/day (45M/month)** on SageMaker at **p95 1.8s** with **99.99% success**, and cut loan-file document verification from about 4 days down to 4–5 hours.
 - Designed the **five-branch multimodal architecture**: ResNet50 plus CNN, BiLSTM, and a **multi-head-attention Transformer I wrote from scratch** over frozen GloVe. Trained it on **Horovod across 20x A10G GPUs**, after fixing the host-RAM OOM and all-reduce failures that were blocking runs. It gets **weighted F1 0.959 to 0.972** across five lender benchmarks.
 - Proposed and built the OCR-free successor** to get OCR out of the inference path: **Donut / Swin encoder, 75M parameters fully fine-tuned**, PyTorch Lightning DDP on **20x L40S** at bf16. It hit **weighted F1 0.975** on 528,574 pages against the OCR-based production ensemble's 0.974. I exported it to **ONNX**, benchmarked 9 SageMaker instance types and picked one **~20% cheaper per hour**.
 - Led the P1 root-cause analysis that cleared the model**. 21,096 pages were reported failed in a 260K-page regression. I traced 5 services in CloudWatch Logs Insights and ruled out both the model and the outage hypotheses. The real cause was in the platform Java path: an input validator was rejecting whole documents before inference, so **one dropped page could null a 530-page file**. True failure rate was **2.3%**, and the platform team shipped the fix I recommended.
 - Recovered a production accuracy regression across **946,355 pages**. Fixed a train/benchmark taxonomy mismatch that **took F1 from 0.62 to 0.92**, then found 64% of what was left was correct predictions being thrown away by one confidence threshold. I own the per-class threshold config — **calibrated selective prediction** across all 643 classes.
 - Shipped **four consecutive production model releases** through the Jenkins-to-SageMaker pipeline: cross-account artifact promotion, model registration, previous-version teardown, and one rollback. Handled autoscaling from **20 to 118 instances** under burst, and took every release from staging to production through change control. Also migrated the serving export from **TF1 to TF2**.

643	1.7M	0.975	99.99%	7.1M	9 yrs
CLASSES IN PRODUCTION	REQUESTS / DAY	WEIGHTED F1, OCR-FREE	SUCCESS RATE	PAGES DEDUPLICATED	DOCUMENT AI

EXPERIENCE — CONTINUED

- Persistent Systems** — Client: ICE Data Services **Aug 2021 – Nov 2023**
Senior ML Engineer
- Built the data-ingestion and OCR layer that the production classifier still runs on. Decrypted lender image sets, extracted text at scale with **Tesseract**, and normalised **27+ source schemas** into one labelled corpus using PySpark UDFs.
 - Added the **OpenCV preprocessing stage** — thresholding, denoising, skew correction, normalisation — that raised OCR quality for every downstream model.
 - Built the first multimodal **CNN + BiLSTM** training pipeline on Databricks, including the BiLSTM encoder-decoder that produces the sequence representations the text branches consume. The production 643-class ensemble was scaled up from this.
 - Set up the initial **SageMaker** deployment and monitoring path plus **MLflow** experiment tracking. The current Jenkins-to-SageMaker release pipeline was built on top of that.
- Zyclyx** — Client: Al Rajhi Bank, Riyadh **Aug 2019 – Mar 2021**
Data Scientist
- Designed and built the bank's document-digitisation pipeline **as its first ML engineer**, running over **~100,000 bilingual Arabic and English documents** end to end — classification, field detection, OCR and entity extraction.
 - Built the model chain: a **ResNet50** classifier over **~65 document types (~88% accuracy)**, TensorFlow Object Detection to localise the required fields, then region cropping, Tesseract OCR, and a **GRU-based NER** to pull out entity values (name, country, ID number). Also built a deep-learning **signature-verification model** to flag forged signatures.
 - Traced the pipeline's largest error source upstream to the branches, each scanning at different settings — colour distortion, rotation and skew. I audited scanner configurations on-site across the bank's major branches and derived a standard profile, and **the bank issued it as a circular mandating those settings bank-wide**. That fixed the problem at the source instead of in preprocessing.
 - Integrated extraction into the bank's RPA (BluePrism) workflow behind a **human-in-the-loop review gate** — every page checked before commit, corrections fed back as training signal. Shipped **on-premise** as Flask + Unicorn microservices, so no document left the bank's network.
 - Mentored 5+ engineers** as the team grew around the pipeline I had built.

- VISSINDIA** — Hyderabad **Feb 2017 – Aug 2019**
Python & Computer Vision Engineer
- Built real-time computer-vision pipelines for identity verification and multi-object tracking: face detection (Haar cascades, HOG, CNN), keypoint matching (SIFT / SURF / FLANN), and TensorFlow object detection across video frames. Also built the OpenCV preprocessing layer that made noisy imagery usable by the detection models.

KEY PROJECTS

- Rulebook-RAG** — Retrieval-Grounded Document Classification · personal project, 2026
- Built a **36-class** document classification system with **no trained model in the loop and no labelled training data**. Class knowledge sits in a written rulebook generated from official instruction documents, retrieved per query. To add a class I write one rule and rebuild the index: **0.83 accuracy** on 5 classes added after the index was built, with no retraining.
 - Hand-wrote Okapi BM25** and fused it with dense embeddings using reciprocal rank fusion (**+5.2 pts**), then added cross-encoder reranking (**+3.9 pts**). Every prediction cites a rule id, resolved against the live index before the answer returns: **0 invalid of 775**.
- Document Deduplication Pipeline** · ICE Data Services
- Built near-duplicate detection across a **7.1M-page** mortgage corpus. Generated DonutSwin page embeddings with **TorchDistributor across 40 GPU nodes**, then ran a two-tier **FAISS IndexFlatIP** cosine search — one pass at page level, one on mean-pooled document embeddings.
 - Ran the searches inside label and page-count groups **so I could use many small exact searches instead of one approximate index**, and used **Union-Find** to collapse transitive duplicate chains. Flagged **67% of pages / 69% of documents** as near-duplicates.